



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination
Academic Year: 2025-26 (semester I)

Class: Final Year B. Tech.

Course: Sustainable Chemical Technology

Course Code:
PEC- CH409

Date: 18-08-2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. All questions are compulsory.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Figures to right indicate full marks.
- iv. The symbols carry usual meaning.

Q.N.		Marks	CO	BT
Q.1	a	3	CO1	BT1
	b	3	CO1	BT1
	c	4	CO1	BT3
Q.4	a	3	CO2	BT2
	b	3	CO2	BT3
	c	4	CO2	BT2

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the different types of environmental pollution problems and their sustainable solutions.
CO 2	Analyse the principles of sustainability.
CO 3	Work in the area of sustainability for research and education
CO 4	Environmental Impact Assessment Studies Sustainable Habitat
CO 5	Resources and its utilisation.



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Department of Chemical Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech.

Course: Catalytic Science and Engineering

Course Code: PEC-CH408

Date: 18.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. *All questions are compulsory.*
- ii. *Assume suitable data whenever necessary and state it clearly.*
- iii. *Figures to right indicate full marks.*

Q.N.			Marks	CO	BT
Q.1	a	Define heterogeneous catalysis. Provide an example of an industrial process that utilizes a heterogeneous catalyst.	3	CO1	BT1
	b	Compare physisorption and chemisorption in detail	3	CO1	BT4
	c	Discuss the Sonochemical methods for preparing supported heterogeneous catalysts briefly.	4	CO2	BT3
Q.2	a	3g of activated charcoal was added to 50 ml of acetic acid solution (0.06 N) in a flask. After an hour, it was filtered, and the strength of the filtrate was found to be 0.042 N. Calculate the amount of acetic acid adsorbed per gram of charcoal?	3	CO2	BT5
	b	Write a short note on Zeolite catalyst	3	CO2	BT2
	c	Describe the construction and working of FTIR spectroscopy.	4	CO3	BT4

=== ALL THE BEST ===

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply knowledge of heterogeneous catalytic reactions in industrial applications.
CO 2	Find out the mechanism and kinetics of heterogeneous catalytic reactions
CO 3	Prepare the required catalyst and carry out characterization of the same
CO 4	Predict mass and heat transfer effects on heterogeneous catalytic reactions
CO 5	Design reactors for heterogeneous catalytic reactions.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
ISE-I- Examination (Semester-I) 2025-26

Department : Chemical Engineering

Class : Final Year B.Tech

Sub: Piping Engineering

Sub.Code: PEC-CH405

Date: 18/08/2025

Time: 11:00 - 11:45

Max. Marks :20

Note: i. Attempt all questions.

ii. Assume suitable data if necessary

iii. Use of non-programmable calculator is permitted

Q.N.		Questions	Marks	CO	BT
Q. 1	a	What do you understand by piping engineering? Give the role, responsibilities of Piping Engineering	04	1	2
	b	What are the material selection criteria for piping system?	03	1	2
	c	What do you understand by schedule number? Which are the factors are affected on fluid flow in pipes?	03	2	2&3
		OR			
	c	Calculate allowable internal pressure P for schedule 40 mild steel pipe having ultimate tensile strength (S value) of 65300 psi.	03	2	5
Q. 2	a	What is significance of codes? Explain in brief, codes for pipes used in chemical process industries.	04	2	2&3
	b	What do you understand by Newtonian and non-Newtonian fluids?	02	1	
	c	Water flows a pipe 25 mm in diameter at a velocity of 6 m s^{-1} . Determine whether the flow is laminar or turbulent. Assume that the dynamic viscosity of water is $1.30 \times 10^{-3} \text{ kg m}^{-1} \text{ s}^{-1}$ and its density is 1000 kg m^{-3} . If oil of specific gravity 0.9 and dynamic viscosity $9.6 \times 10^{-2} \text{ kg m}^{-1} \text{ s}^{-1}$ is pumped through the same pipe, what type of flow will occur.	04	2	5

Course Outcomes (CO)

- CO-1 Capable to understand the concepts of the nature of fluid flow
- CO-2 Ability to analyze Flow of liquid in Pipe.
- CO-3 Should understand the concept of Pipe surface roughness effects.
- CO-4 Get familiar with phenomenon of pipe networking.
- CO-5 Understand the difference between pipes and tubes.
- CO-6 Understand the Apparent Reynolds.

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Department of Chemical Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: Final Year B. Tech

Course: Industrial Organization
and Management

Course Code: HMC-CH404

Date: 19.08.2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. Solve any TWO questions.
- ii. Figures to right indicate full marks.

Q.N.			Marks	CO	BT
Q.1	a	What are the role and responsibilities of management?	05	CO1	BT2
	b	Explain the difference and relationship between management, administration and organisation.	05	CO1	BT3
Q.2	a	What is management science? Write basic approach of the scientific management.	06	CO2	BT1
	b	Write the skills required for a successful manager.	04	CO1	BT2
Q.3	a	Explain in brief the factors involve in management process.	05	CO2	BT2
	b	Write the contribution of GANTT for scientific management.	05	CO1	BT1

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the management process and structure in the industry so that it will help them to work in a better way
CO 2	Develop an efficient methodology for industrial management
CO 3	Cater the issues related to current industrial amendments.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Chemical Engineering
In-Semester Examination I
Academic Year: 2025-26 (semester I)

Class: Final Year B.Tech (Chemical)	Course: Process Modelling and Simulation	Course Code: PCC-CH402
Date: 21.08.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note: All questions are compulsory and figures to right indicate full marks
State suitable assumption whenever necessary and state it clearly.

Q.N.	Question	Marks	CO	BT
Q.1	(a) Use dimensionless modelling approach to develop mathematical model for first order isothermal reaction carried out in the CSTR	5	2	5
	(b) Compare empirical approach verses process modelling and simulation approach with an example. Draw schematic block diagram for the proper representation	5	1	4
Q.2	(a) Write the component continuity equations for a perfectly mixed batch reactor (no inflow or outflow) with first-order isothermal reactions: i. Consecutive ii. Simultaneous iii. Reversible	5	1	3
OR				
Q.2	(a) Apply all the fundamentals to convert accumulation term of component balance equation into mathematical form for a process which involves gaseous constituents.	5	1	5
Q.3	(a) State and write a general energy balance equation used in the chemical engineering process modelling. Draw information flow diagram for represent interaction and information sharing in the energy balancing	5	1	2&3
	OR			
	(a) Formulate the model equations and simulate it for a solid solute getting leached in batch extraction vessel.	5	3	6

=== ALL THE BEST ===

Course Outcomes (CO)

CH402.1 Able to apply fundamental laws & principles to formulate process models

CH402.2 Able to develop process model systems for chemical processes.

CH402.3 Able to select proper modeling system and find simulated results for chemical processes.

CH402.4 Able to apply and solve chemical processes using commercial software packages.

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